

Data, Information, & Knowledge

in the face of the COVID-19 pandemic

Dr. Ata Kalirad

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سلسله سخنرانی های عمومی COVID-19

یکشنبه ۱۷ فروردین ۱۳۹۹

COVID-19 Overview: Common questions and concerns

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پنجشنبه ۲۱ فروردین ۱۳۹۹

Data, information, and knowledge in the face of the COVID-19 pandemic

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پنجشنبه ۲۸ فروردین ۱۳۹۹

Mathematical models prescribe:

Staying at home is the key to control the spread of COVID-19

Zahra Eidi, PhD, *School of Biological Sciences (IPM)*



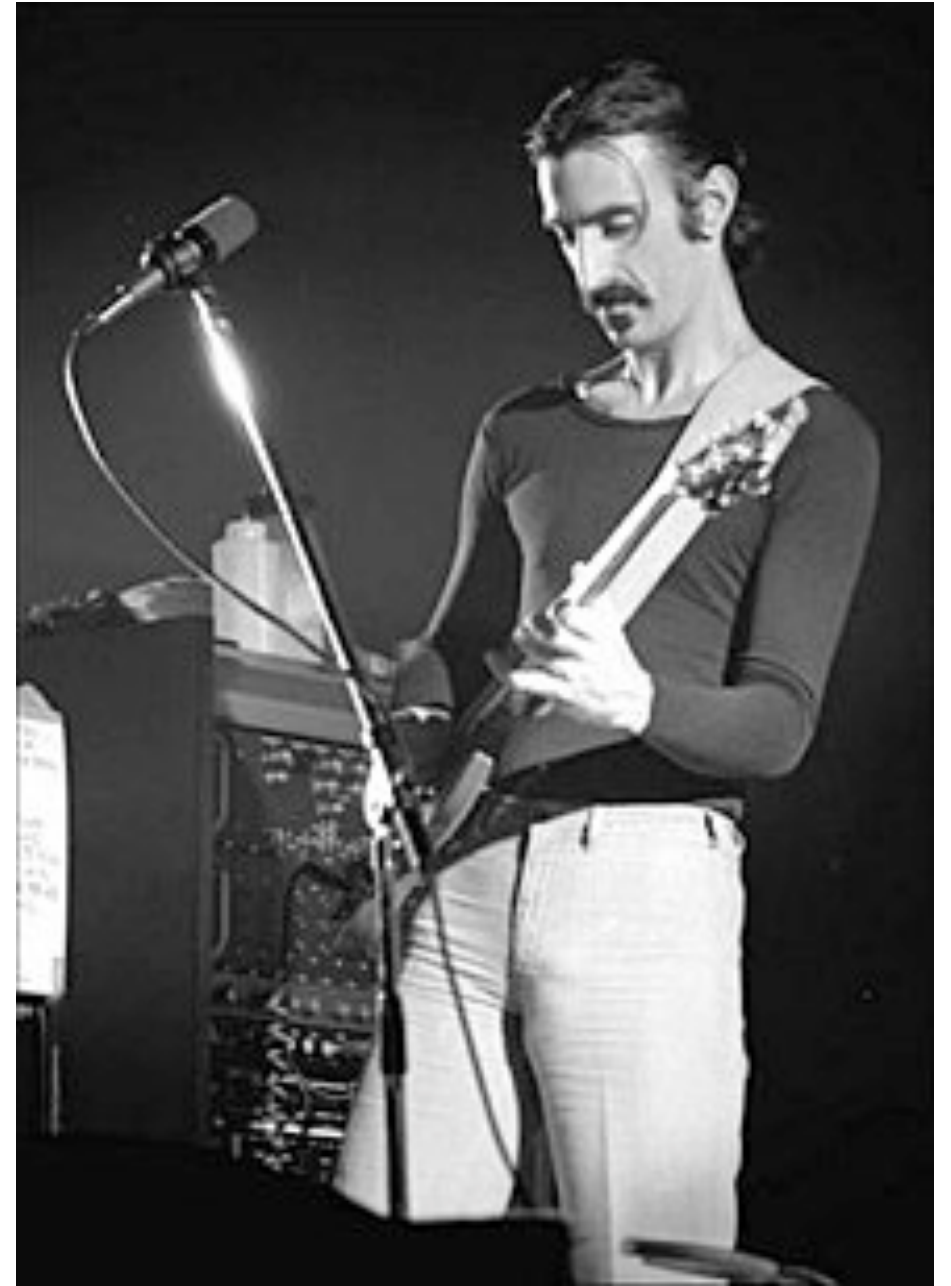
سخنرانی ها ساعت ۱۱ صبح آغاز و از طریق لینک زیر در دسترس خواهند بود.

<https://evnd.co/1GtIh>

شرکت برای عموم رایگان می باشد.

Timeless Wisdom - 1

Data is not **information**,
information is not
knowledge, **knowledge**
is not **wisdom**, **wisdom**
is not **truth**.



Royar (1994) paraphrasing Frank Zappa's (1979) anadiplosis



Part I: What we know

The emergence of infectious disease

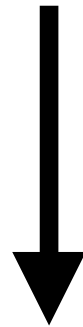
Examples of human infectious diseases of animal origin,
and apparent date(s) of species crossover

Disease	Microbe	Animal source	Date of crossover
Malaria	Parasite	Chimpanzee	ca. 8000 BCE
Measles	Virus	Sheep or goat	ca. 8000 BCE
Smallpox	Virus	Ruminant?	>2000 BCE
Tuberculosis	Mycobacterium	Ruminant?	>1000 BCE
Typhus	Rickettsia	Rodent	430 BCE and 1492 CE ^a
Plague	Bacterium	Rodent	541 CE, 1347 CE and 1665 CE
Dengue	Virus	Monkey	ca. 1000 CE
Yellow fever	Virus	Monkey	1641 CE
Spanish flu	Virus	Bird, pig	1918 CE
AIDS/HIV-1	Virus	Chimpanzee	ca. 1931 CE
AIDS/HIV-2	Virus	Monkey	Twentieth century

McLean et al. (2005) *SARS-A Case Study in Emerging Infections*

The role of adaptation

**more genetically
variability**



**Better chance to
adapt to replicate
in the new host**

Error-threshold quandary

Copying fidelity

$$0 < q < 1$$

Fidelity for sequence of N

$$Q(N) = q^N$$

Error-threshold quandary

Copying fidelity

$$0 < q < 1$$

Fidelity for sequence of N

$$Q(N) = q^N$$

For a system with sequence m and j (fixed total number):

$$\dot{x}_m = A_m \times Q_m \times x_m - x_m \times C$$

$$\dot{x}_j = A_j \times Q_j \times x_j - x_j \times C$$

Error-threshold quandary

$$\dot{x}_m = A_m \times Q_m \times x_m - x_m \times C$$

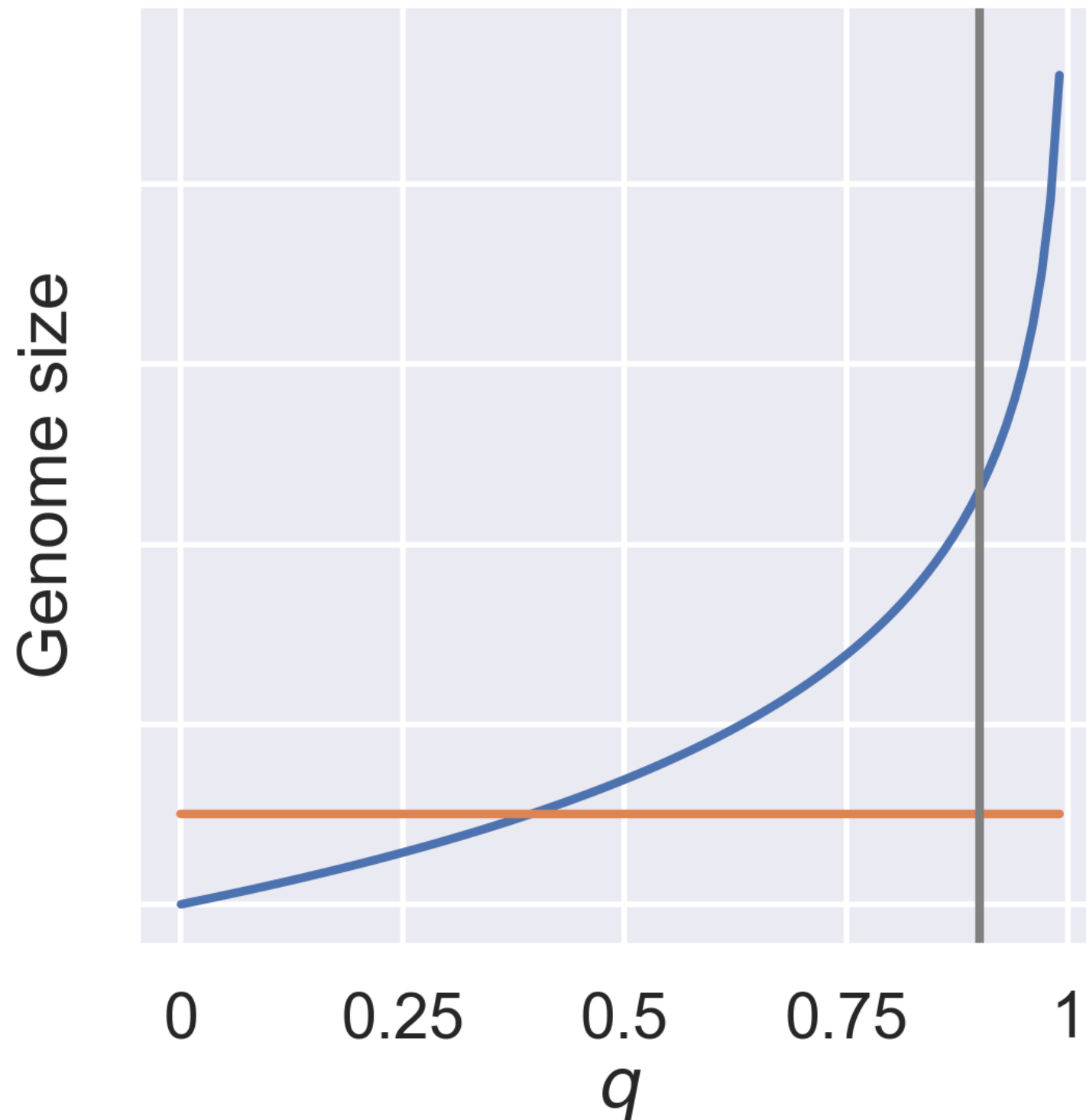
$$\dot{x}_j = A_j \times Q_j \times x_j - x_j \times C$$

At equilibrium with Non-zero master sequences:

$$Q > \frac{A_j}{A_m} = \frac{1}{s} \quad , \quad Q = q^N \approx e^{-N(1-q)}$$

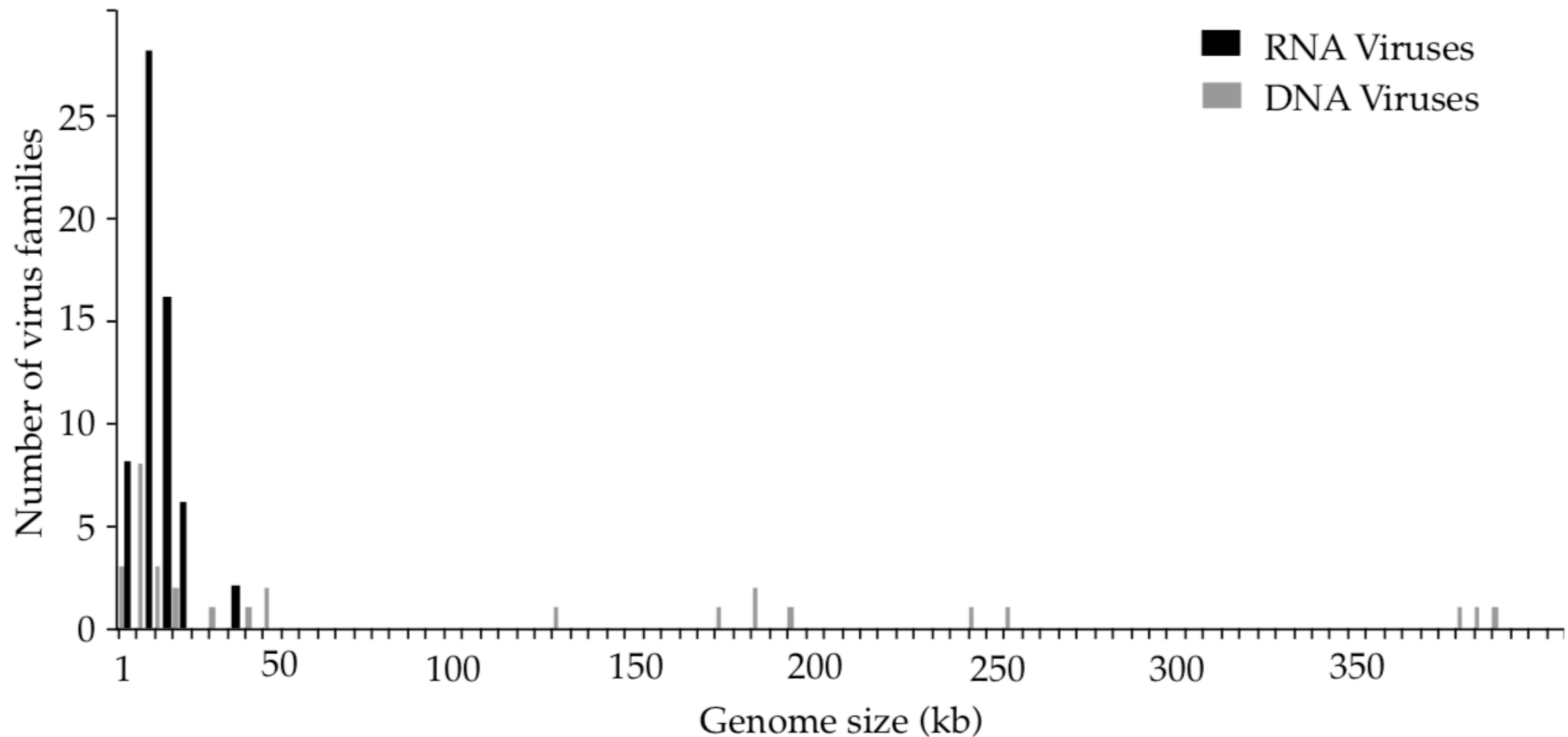
Error threshold =>
$$N < \ln s / (1 - q)$$

Error-threshold quandary



Maynard Smith & Szathmary (1997) *The Major Transitions in Evolution*

Error-threshold limits RNA virus genome size

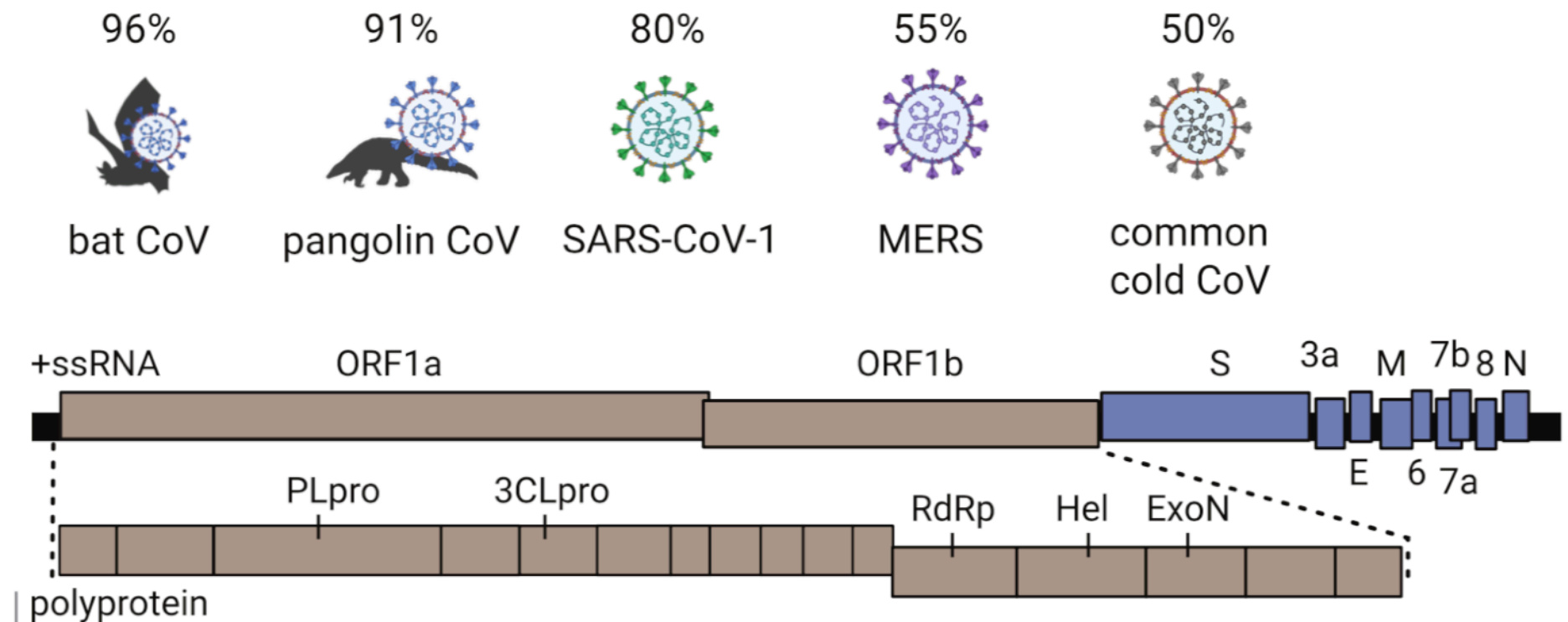


McLean et al. (2005) *SARS-A Case Study in Emerging Infections*

Is SARS-CoV-2 mutating faster than usual?

Genome

Nucleotide identity to SARS-CoV-2



Length: $\approx 30\text{kb}$; β -coronavirus with 10-14 ORFs (24-27 proteins)

Evolution rate: $\sim 10^{-3} \text{ nt}^{-1} \text{ yr}^{-1}$ (measured for SARS-CoV-1)

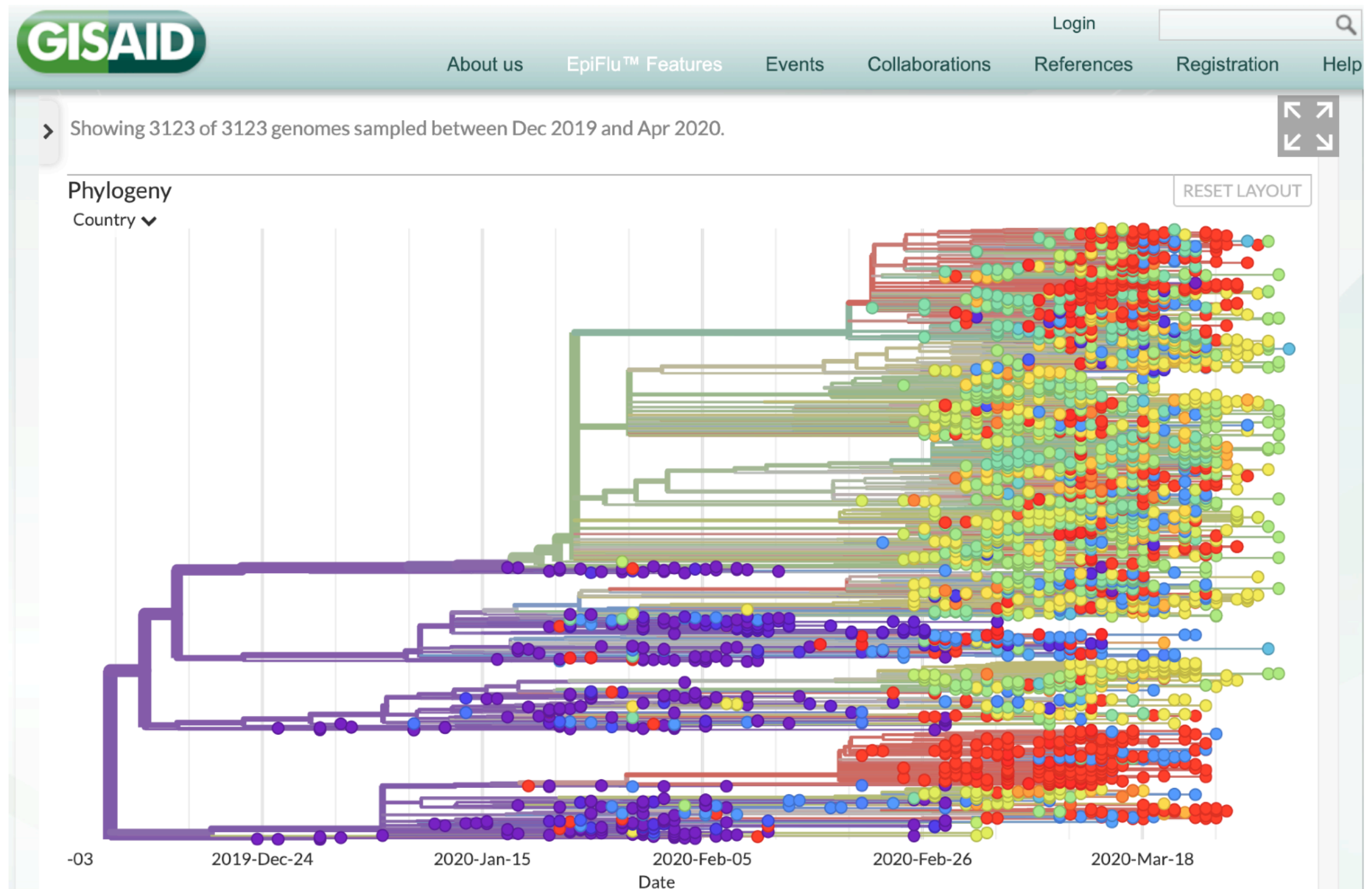
Mutation rate: $\sim 10^{-6} \text{ nt}^{-1} \text{ cycle}^{-1}$ (measured for MHV coronavirus)

Bar-On et al. (2020) *SARS-CoV-2 (COVID-19) by the numbers*

Is SARS-CoV-2 mutating faster than usual?

	Genome size	Mutation rate per site per cycle
Influenza virus A	13588	1.5×10^{-5}
HIV Type I	9181	4.9×10^{-5}
Measles virus	15894	4.4×10^{-5}

The phylogeny of SARS-CoV-2





Part II: The known unknowns

Timeless Wisdom - 2

“There are **known knowns**; there are things we know we know. We also know there are **known unknowns**; that is to say we know there are some things we do not know.”



Donald Rumsfeld, U.S. Department of Defense (DoD) news briefing on February 12, 2002

The problem with prediction

Risk: Something that you can put a price on. It is not pleasant when you take a “bad beat” in poker, but at least you know the odds of it and can account for it ahead of time.

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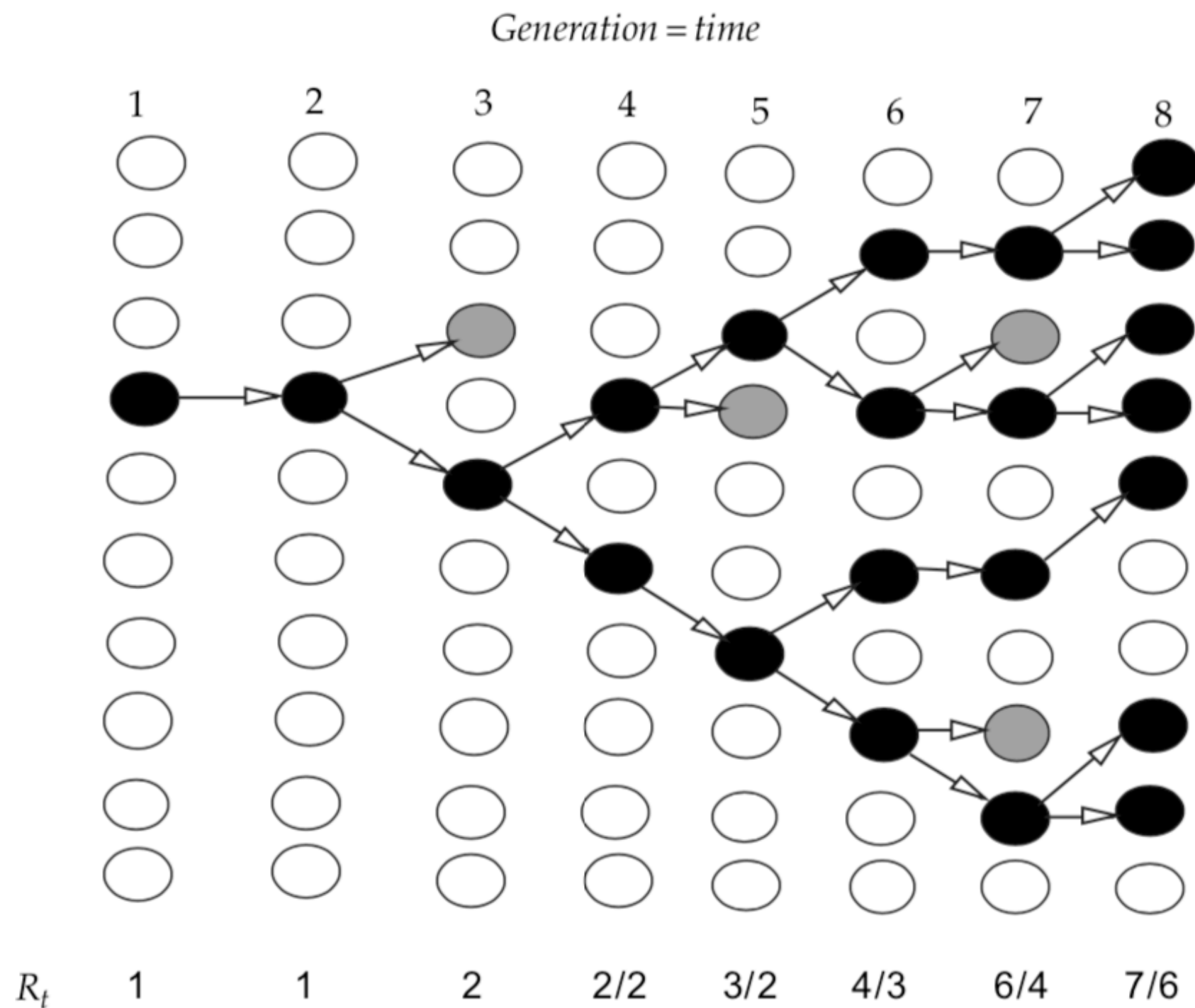
Uncertainty: Risk that is hard to measure. You might have some vague awareness of the demons lurking out there. You might even be acutely concerned about them. But you have no real idea how many of them there are or when they might strike.

What is a good prediction?

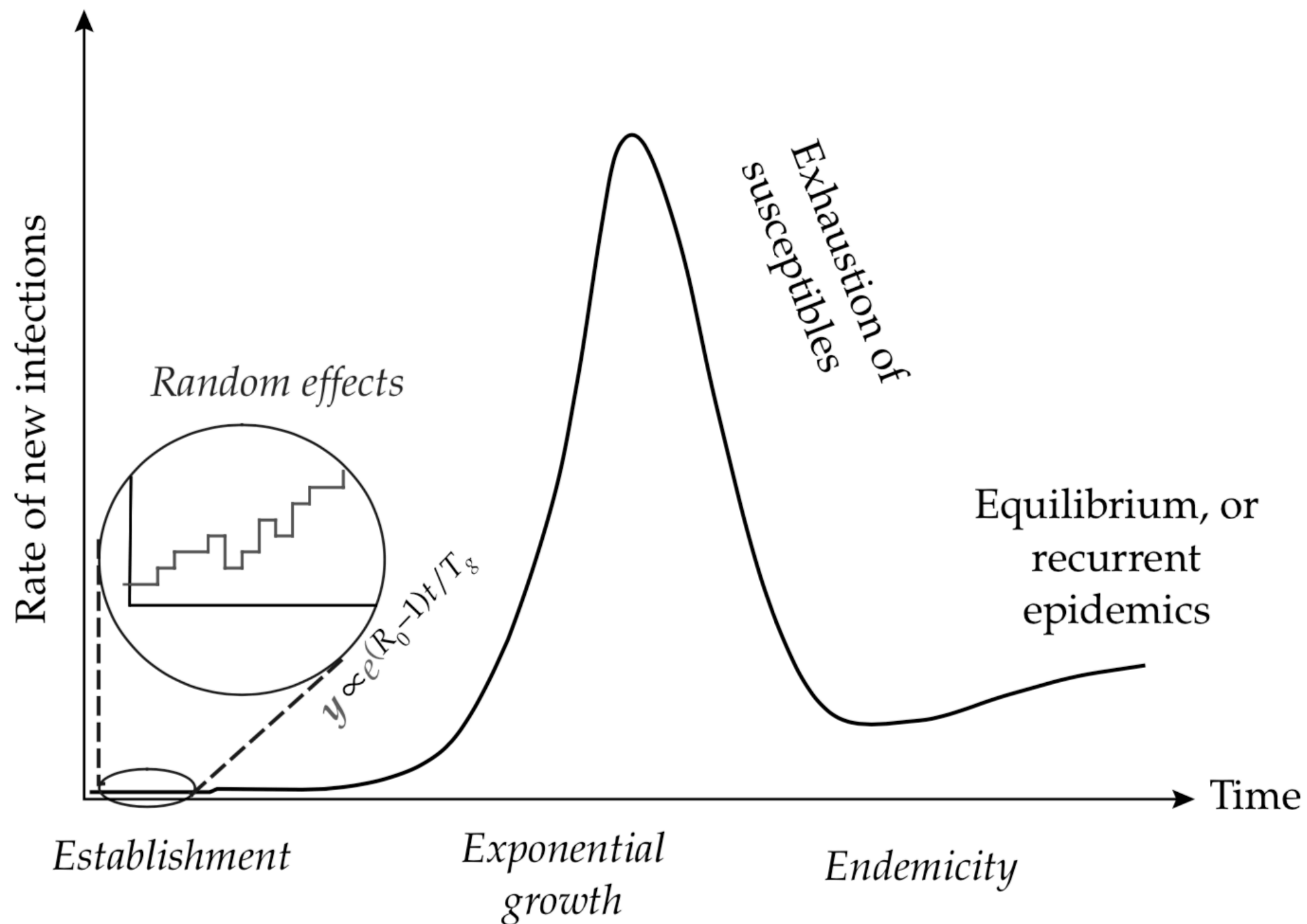
Consistency	correspondence between forecasts and judgment
Quality	correspondence between forecasts and observation
Value	Incremental benefits of forecast to users

Allan H. Murphy, "What Is a Good Forecast? An Essay on the Nature of Goodness in Weather Forecasting," AMS (June 1993)

From general models to specific prediction

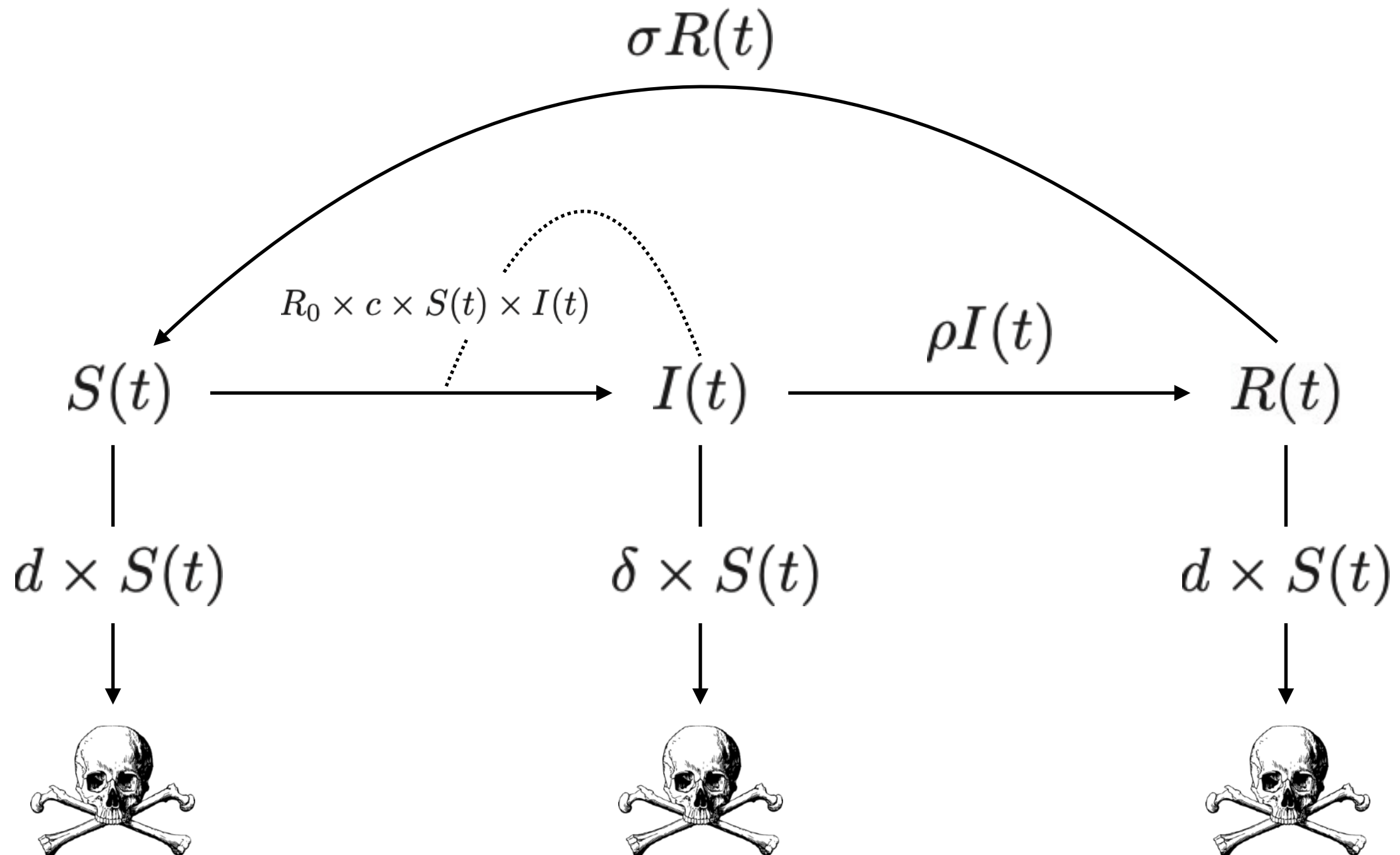


From general models to specific prediction



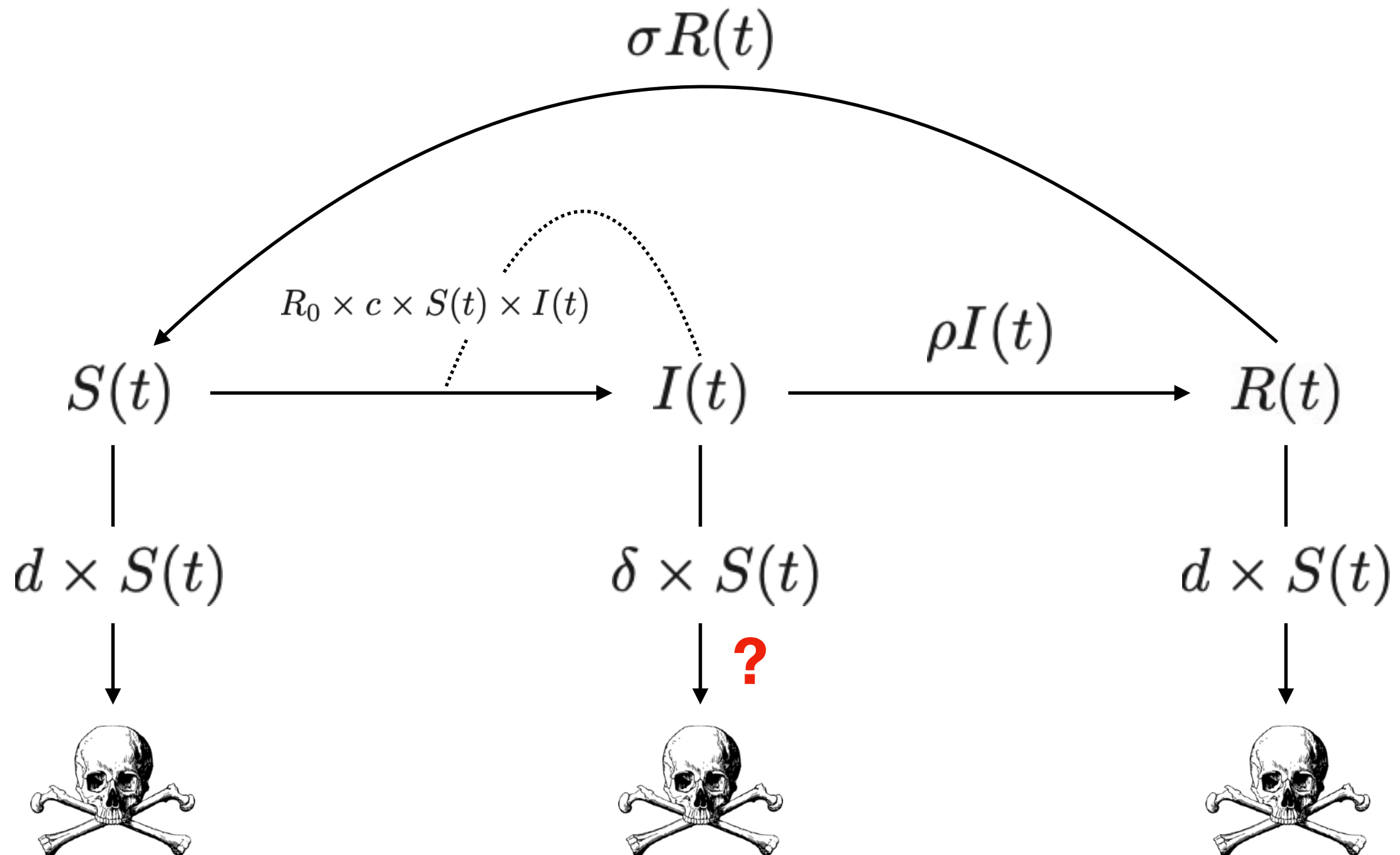
From general models to specific prediction

SIR model



From general models to specific prediction

SIR model



From general models to specific prediction

$$\text{Case fatality rate (CFR)} = \frac{\text{Number of death due to the disease}}{\text{Total number of cases}}$$

From general models to specific prediction

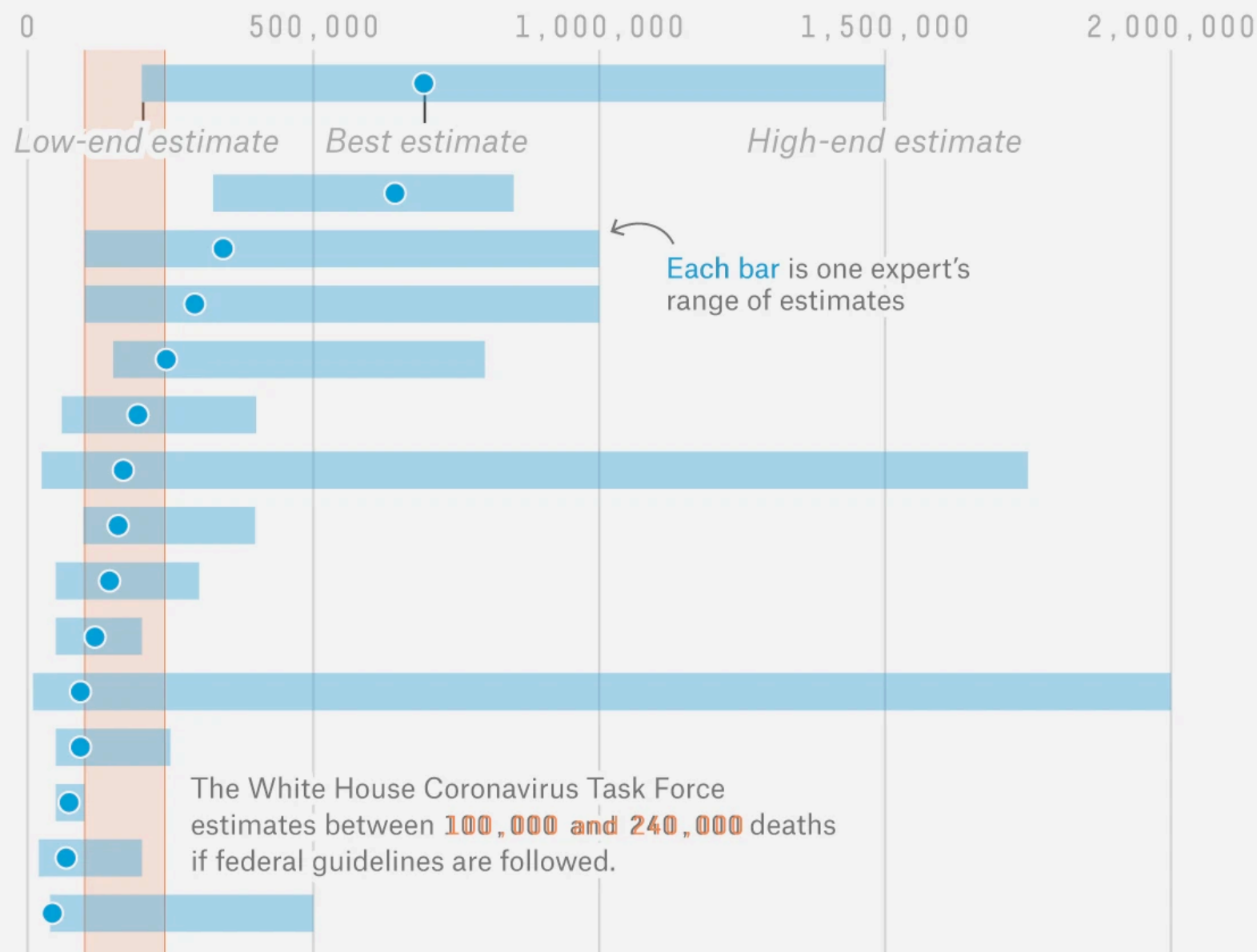
$$\text{Case fatality rate (CFR)} = \frac{\text{Number of death due to the disease}}{\text{Total number of cases}}$$

$$\text{Case fatality rate (CFR)} = \frac{\text{Varied}}{\text{Unknown}}$$

From general models to specific prediction

Still a lot of uncertainty around the coronavirus death toll

Expert estimates of the total number of COVID-19 deaths in the U.S. in 2020

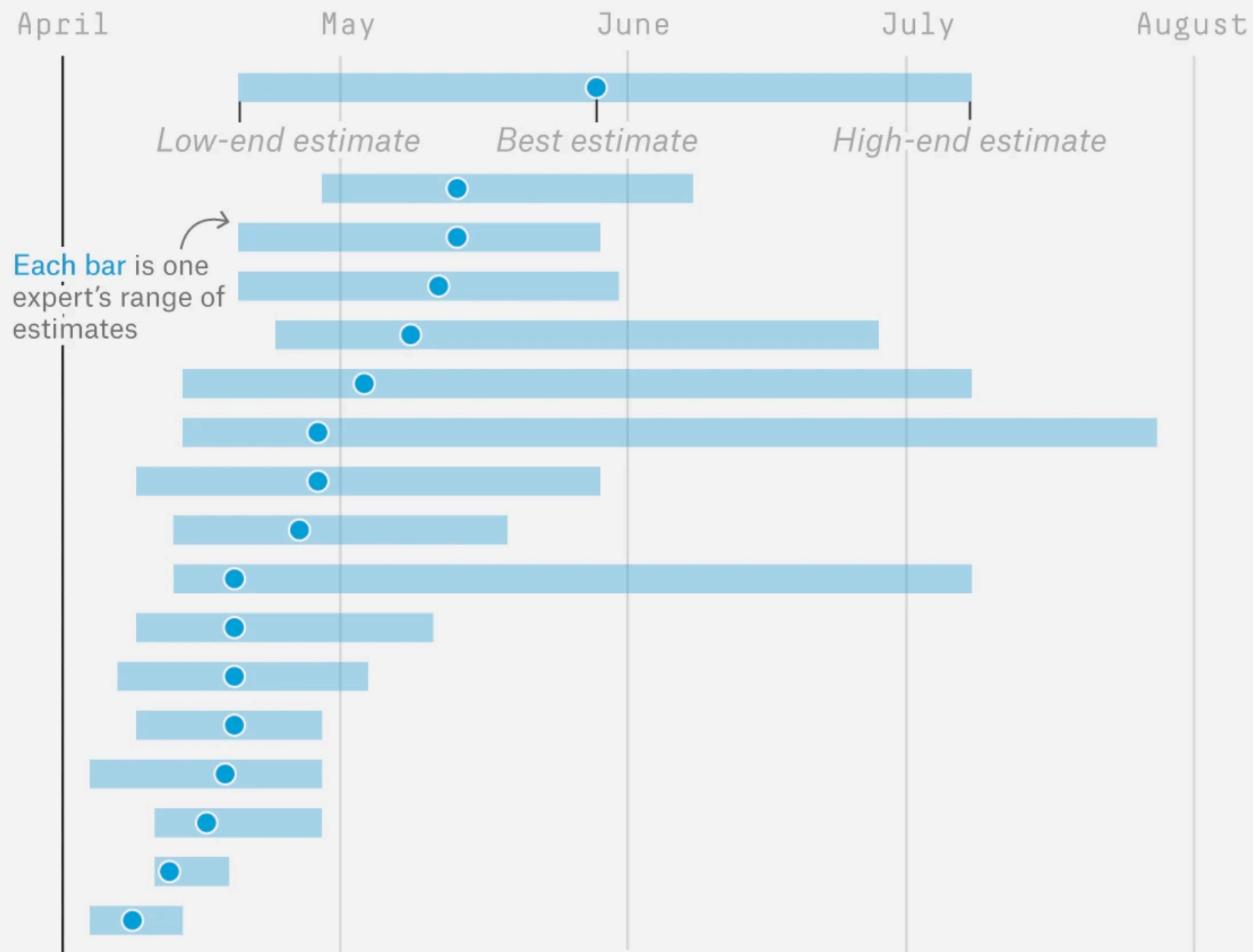


Responses are from a survey of experts conducted March 30-31. Chart omits one expert whose best estimate was 1 million and high-end estimate was 10 million.

From general models to specific prediction

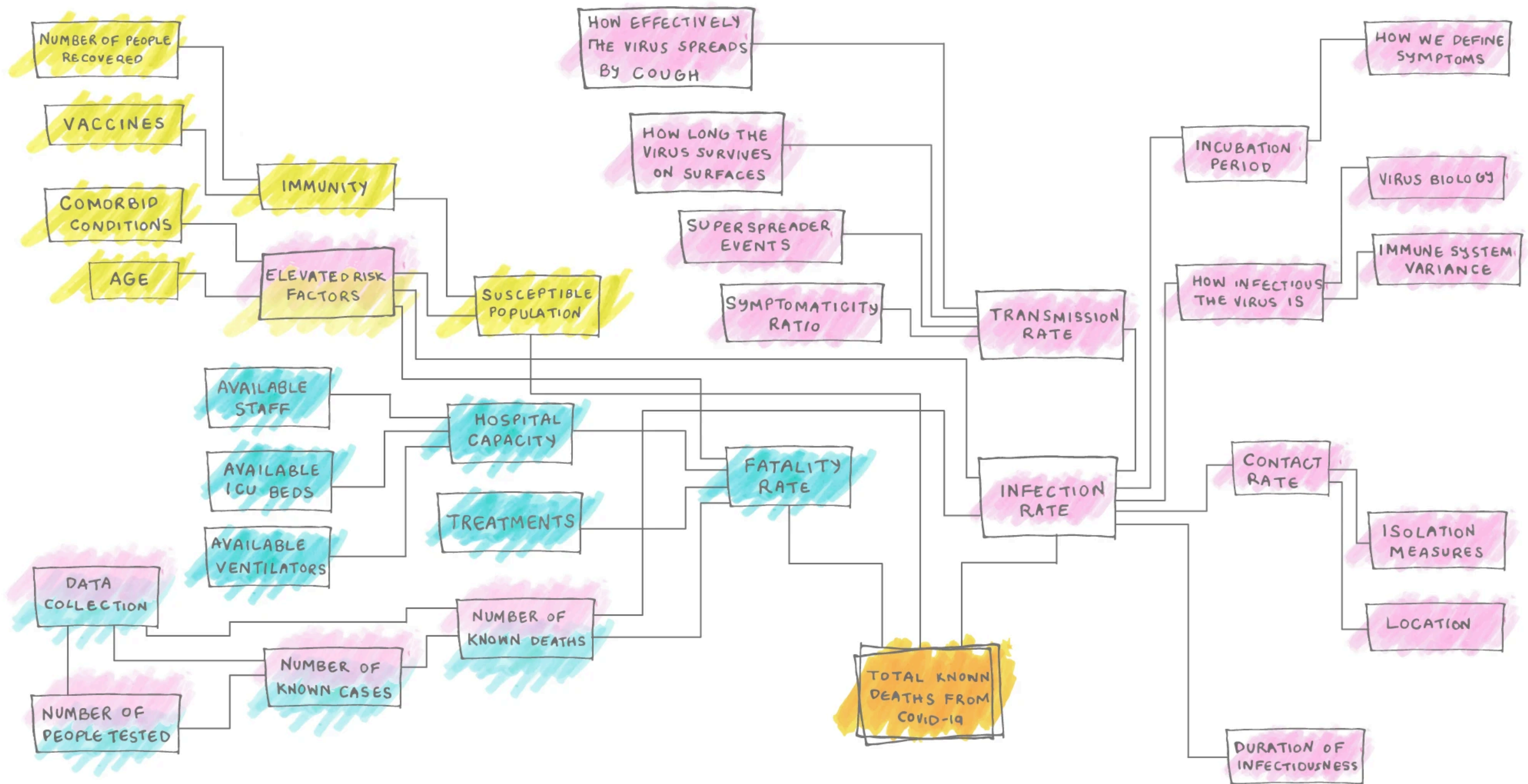
New York state may have a long road ahead

Expert estimates of when New York state's daily COVID-19 cases will drop below 1,000



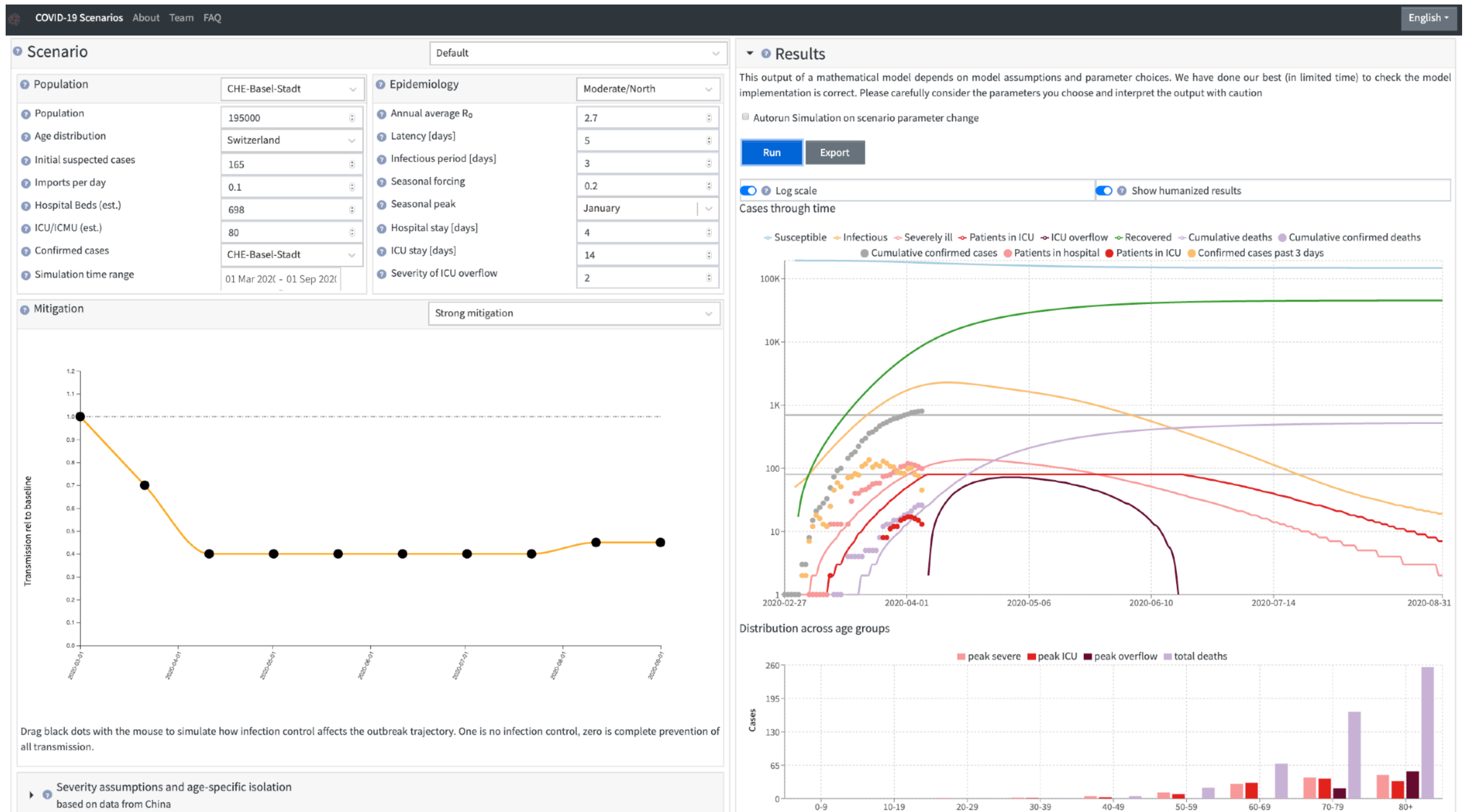
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From general models to specific prediction



Why It's So Freaking Hard To Make A Good COVID-19 Model from <https://fivethirtyeight.com/>

From general models to specific prediction



From *neherlab.org*

What shall we do?

“The global scientific assault on COVID-19 is faster [...] I am hopeful that science will deliver on this challenge, too. But I worry that engendering false hope will cause complacency that will deprive us of the time needed to find a lasting solution. And I worry about lasting damage if science overpromises.

Let's underpromise. Let's overdeliver.”



H. Holden Thorp, Science 27 Mar 2020



Fin.